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corrp: An R package for multiple correlation-like analysis and clustering in mixed data

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Summary

Correlation-like analysis provides an important statistical measure that can describe the size and direction of an association between variables. However, there are few R packages that can efficiently perform this type of analysis on large datasets with mixed data types. The `corrp` package provides a full suite of solutions for computing various kinds of correlation-like measures such as Pearson correlation (Pearson, 1895), Distance Correlation (Székely et al., 2007), Maximal Information Coefficient (MIC) (Reshef et al., 2011), Predictive Power Score (PPS) (Wetschoreck et al., 2020), Cramér's V (Cramér, 1946), and the Uncertainty Coefficient (Theil, 1972). These methods support the analysis of data frames with mixed classes (integer, numeric, factor, and character). In addition, it offers a C++ implementation of the Average Correlation Clustering Algorithm (ACCA) (Bhattacharya & De, 2010), which is original used in genetic studies. In this package, the ACCA algorithm has been extended to work directly with correlation matrices derived from different association methods, depending on the data types and user preferences. Furthermore, the package is designed for parallel processing in R, making it highly efficient for large datasets.

Statement of need

The corrp package is an R package that provides a flexible and efficient way of doing correlation-like analysis on mixed-type data frames. Most traditional correlation methods in R are applicable only to specific data types or small datasets `corrp` extends this capability by handling mixed classes, integrating various association methods, and offering clustering directly from the resulting correlation matrix. The package is particularly useful for researchers and data scientists working with complex datasets who require robust and scalable tools for both association analysis and clustering.

Implementation

The `corrp` package integrates R and C++ to combine the flexibility of R with the speed of C++, optimizing key operations. Its core functionalities include the selection of correlation-like methods based on pair of variable types (numeric pairs, numeric and categorical pairs, etc.). Users can create correlation matrices, remove variables based on significance, and cluster the correlation matrix using the ACCA clustering algorithm. This approach has been modified to support mixed data types and various correlation methods. Additionally, the package supports parallel processing through the `foreach` package, significantly improving performance on large datasets.

As mentioned before, one can choose between the following options based on the type pair:

- **Numeric pairs (integer/numeric):**

  - Pearson correlation coefficient (Pearson, 1895), a widely used measure of the strength and direction of linear relationships.
  - Distance Correlation or distance covariance (Székely et al., 2007), based on the idea of expanding covariance to distances, can measure both linear and nonlinear associations between variables.
  - Maximal Information Coefficient (MIC) (Reshef et al., 2011), a information-based nonparametric based method that can detect linear or non-linear relationships between variables.
  - Predictive Power Score (PPS) (Wetschoreck et al., 2020), a metric used to assess predictive relations between variables.
- **Numeric and categorical pairs (integer/numeric - factor/categorical):**
  - Square root of the R^2 coefficient from linear regression (Cohen, 1983).
  - Predictive Power Score (PPS) (Wetschoreck et al., 2020).
- **Categorical pairs (factor/categorical):**
  - Cramér's V (Cramér, 1946), a measure of association between nominal variables.
  - Uncertainty Coefficient (Theil, 1972), a measure of nominal association between two variables.
  - Predictive Power Score (PPS) (Wetschoreck et al., 2020).

Usage

The `corrp` package provides seven main functions for correlation calculations, clustering, and basic data manipulation:

- **corrp**: Performs correlation-like analysis with user-specified methods for numeric, categorical, factor, interger and mixed pairs.
- **corr_matrix**: Generates a correlation matrix from analysis results.
- **corr_rm**: Removes variables based on p-value significance.
- **acca**: Performs the ACCA clustering algorithm with added support for mixed data types.
- **sil_acca**: A C++ implementation of the Silhouette method for interpreting and validating the consistency of clusters within ACCA clusters of data.
- **best_acca**: Determining the optimal number of clusters in ACCA clustering using the average silhouette approach.

We calculate correlations for the *eusilc* dataset using the Maximal Information Coefficient for numeric pairs, Predictive Power Score for numeric/categorical pairs, and Uncertainty Coefficient for categorical pairs. This synthetic dataset represents Austrian EU-SILC data on income, demographics, and household characteristics.

```r
set.seed(2024)
library("laeken")
library("corrp")
data(eusilc)

eusilc = eusilc[, c("eqSS", "eqIncome", "db040", "rb090")]
colnames(eusilc) = c("House_Size", "Income", "State", "Sex")

results = corrp(
  eusilc,
  cor.nn = 'dcor', cor.nc = 'lm', cor.cc = 'pps',
  verbose = FALSE
)

head(results$data)
```

	infer	infer.value	stat	stat.value
1	Distance Correlation	1.000	P-value	0.000
2	Distance Correlation	0.008	P-value	0.000
3	Linear Model	0.146	P-value	3.57e-64
4	Linear Model	0.071	P-value	4.79e-18
5	Distance Correlation	0.008	P-value	0.000
6	Distance Correlation	1.000	P-value	0.000

	isig	msg	varx	vary
1	TRUE		House_Size	House_Size
2	TRUE		House_Size	Income
3	TRUE		House_Size	State
4	TRUE		House_Size	Sex
5	TRUE		Income	House_Size
6	TRUE		Income	Income

When choosing correlation methods, it's important to think about their performance for different pair types. For **numeric pairs**, **Pearson** is the quickest and most efficient option, while the **Maximal Information Coefficient (mic)** is significantly slower, making it less suitable for large datasets. **Distance correlation (dcor)** is a better performer than mic but still not the fastest choice, and **Predictive Power Score (pps)** is efficient but may take longer than Pearson. For **numeric-categorical pairs**, the **linear model (lm)** typically outperforms pps. In **categorical pairs**, **Cramér's V**, **Uncertainty Coefficient (uncoef)**, and **pps** are options, with **uncoef** being the slowest of the three.

As you increase the number of columns, runtime will grow significantly due to the `N * N` scaling, so choose your methods wisely to ensure efficient performance.

Using the previous result, we can create a correlation matrix as follows:

```
m = corr_matrix(results, col = 'infer.value', isig = TRUE)
m
```

	House_Size	Income	State	Sex
House_Size	1.000	0.008	0.146	0.071
Income	0.008	1.000	0.070	0.071
State	0.146	0.070	1.000	0.000
Sex	0.071	0.071	0.000	1.000

```
# attr(,"class")
# [1] "cmatrix" "matrix"
```

Finally, we can cluster the dataset variables using the ACCA algorithm and the correlation matrix. For example, consider clustering into 2 groups (k = 2):

```
acca.res = acca(m, 2)
acca.res
# $cluster1
# [1] "Sex"    "Income"
#
# $cluster2
# [1] "State"      "House_Size"
#
```

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```
# attr(,"class")
# [1] "acca_list" "list"
```

Performance Improvements

When using the `corrp` function with the `dcor` method for numeric pairs (i.e., `cor.nn = "dcor"`), significant improvements in both memory usage and runtime are observed. This is because the `corrp` package uses a C++ implementation of distance correlation (`dcorT_test`), which is more efficient than the `energy::dcorT.test` function from the `energy` package.

For example, using two vector of length 10000 and 20000, the benchmarks show the following improvements:

Method	10,000		20,000	
	Memory (MB)	Time (sec)	Memory (MB)	Time (sec)
dcorT_test (C++)	4701.44	6.022	18440.54	25.977
energy::dcorT.test	7000.65	13.846	27598.38	60.264

This highlights a substantial reduction in both memory usage and execution time, making the `corrp` package more scalable for larger datasets when applying distance correlation methods. The memory reduction is particularly important because calculating distance correlation requires constructing a distance matrix of size N^2, where N is the length of the input vector. As N grows, the memory demands can quickly become prohibitive.

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